

DarkSide Ransomware – Practical III
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CAP6137 – Malware Reverse Engineering

Executive Summary

The sample is a ransomware created by adversaries who identify themselves as “DarkSide”. It encrypts all the files on the victim’s system, and leaves a readme file. The file contains a tor link to their website <http://darksidefqzcuhtk2.onion/> and ask the victim to download their tool ‘universal decryptor’ to decrypt the file, and have asked them to also paste the provided key on their website.

They have used VMProtect, a popular obfuscator used by even legit companies, especially video game companies, to prevent reverse engineering. The main sample only includes three imports (LoadLibraryA, ExitProcess, and GetProcAddress) responsible for loading other dlls from memory (mentioned in dynamic analysis). It deletes multiple ControlSet registry keys and values, and adds its own class’s keys and values. It makes DNS requests to securebestapp20.com and temisleyes.com and communicates with them over TCP and HTTP.

Static Analysis

subsystem	GUI
compiler-stamp	0x5FE377D3 (Wed Dec 23 12:01:07 2020)
debugger-stamp	0x5FE377D3 (Wed Dec 23 12:01:07 2020)

This sample was compiled on **December 23, 2020**.

Win32VersionValue	0x00000000
subsystem	GUI

This sample is a **GUI** program.

size-without-overlay	n/a
entropy	6.256
imphash	17A48D9C95F7

property	value	value	value	value
name	.text	.rdata	.data	.reloc
md5	52C7889BA37F3177EFE8F5A...	0B8F292A6DD252797194DE2...	C033C139FE22AEE5A9A7633...	BA68B2ECA00E5A89428D63...
entropy	6.269	3.014	5.673	6.501
file-ratio (98.31%)	48.31 %	0.85 %	44.07 %	5.08 %
raw-address	0x00000400	0x00007600	0x00007800	0x0000E000
raw-size (59392 bytes)	0x00007200 (29184 bytes)	0x00000200 (512 bytes)	0x00006800 (26624 bytes)	0x0000C000 (3072 bytes)
virtual-address	0x00401000	0x00409000	0x0040A000	0x00413000
virtual-size (66177 bytes)	0x000071D3 (29139 bytes)	0x00000176 (374 bytes)	0x0000844C (33868 bytes)	0x00000AEC (2796 bytes)
entry-point	0x000081B5	-	-	-
characteristics	0x60000020	0x40000040	0xC0000040	0x42000040

Overall entropy of 6.256 and each section also having entropy less than 7 suggest that this sample is likely not packed. HOWEVER,

name (3)	group (2)	type (1)	ordinal (0)	blacklist (0)	anti-debug (0)	undocumented (0)	deprecated (0)	library (1)
ExitProcess	execution	implicit	-	-	-	-	-	kernel32.dll
GetProcAddress	dynamic-library	implicit	-	-	-	-	-	kernel32.dll
LoadLibraryA	dynamic-library	implicit	-	-	-	-	-	kernel32.dll

There are only 3 imports in this sample, which is a very good indication of it being obfuscated.

ascii	4	0x000000754	-	-	-	UVWS
ascii	15	0x00007265	-	-	-	VMProtect begin
ascii	13	0x000072E2	-	-	-	VMProtect end
ascii	4	0x00009892	-	-	-	W/IN

Also, strings mentioning **VMProtect** indicate that VMProtect was used for obfuscating the program. VMProtect is a common tool used by companies for legitimate software like video games to prevent cracking.

The different sections are found to be:

1. .text: Highest entropy of all sections. This section usually contains the actual code but might also contain the payload in this case.
2. .rdata: Used mainly for importing the needed libraries.
3. .data: Contains global and static variables.
4. .reloc: Stores relocation data for ASLR. Relatively higher entropy suggests that this part might be tampered to conceal malicious payload.

There are only three functions imported as show in the screenshot above. These three are not directly suspicious, but can be used to load suspicious libraries and the functions within them.

type (4)	size (bytes)	file-offset	blacklist (0)	hint (3)	group (4)	value (4/0)
ascii	64	0x0000DDF4	-	size	-	ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz0123456789+/-
ascii	40	0x00000000	-	size-macros	-	!This program cannot be run in DOS mode

Base64 character set is present as a string. Parts of the file are likely to be encoded using base64.

Other important strings, as previously mentioned, are ‘**VMProtect begin**’ and ‘**VMProtect end**’ which indicate the use of VMProtect for obfuscation.

(PS: The screenshots are from different runs and different Operating Systems, therefore there might be some inter-screenshot inconsistencies).

[illegible]

Keys related to 7a1fcd2 ('88c86038' in windows 7) were added.

```
Keys added: 11
-----
HKLM\SOFTWARE\Classes\7a1fcd2
HKLM\SOFTWARE\Classes\7a1fcd2
HKLM\SOFTWARE\Classes\7a1fcd2\DefaultIcon
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Group Policy\ServiceInstances
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Group Policy\ServiceInstances\776594de-cde1-46ce-a4c1-c33f83aa7ee8
HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Group Policy\ServiceInstances
HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Group Policy\ServiceInstances\776594de-cde1-46ce-a4c1-c33f83aa7ee8
HKUS-1-5-21-2013161036-436689623-2610530792-1001\Software\Microsoft\Windows\CurrentVersion\Notifications\Settings\Microsoft.Explorer.Notification.[8B5EAC42-9897-2086-E282-51D34982E322]
HKUS-1-5-21-2013161036-436689623-2610530792-1001\Software\Microsoft\Windows\CurrentVersion\Notifications\Backup\Microsoft.Explorer.Notification.[8B5EAC42-9897-2086-E282-51D34982E322]
HKUS-1-5-21-2013161036-436689623-2610530792-1001\Software\Classes\Local Settings\Software\Microsoft\Windows\CurrentVersion\TrayNotify\ToastAppIdentities\Microsoft.Explorer.Notification.[8B5EAC42-9897-2086-E282-51D34982E322]
HKUS-1-5-21-2013161036-436689623-2610530792-1001\Classes\Local Settings\Software\Microsoft\Windows\CurrentVersion\TrayNotify\ToastAppIdentities\Microsoft.Explorer.Notification.[8B5EAC42-9897-2086-E282-51D34982E322]
```

Certain 7a1fcd2.ico was set as default icon for class 7a1fcd2, and a certain img0.jpg was set as desktop wallpaper

```
Values added: 21
-----
HKLM\SOFTWARE\Classes\7a1fcd2\DefaultIcon: "C:\Users\malware\AppData\Local\7a1fcd2.ico"
HKU\DEFAULT\Software\Classes\Local Settings\Software\Microsoft\Windows\CurrentVersion\Explorer\Background\Wallpapers\Background\img0.jpg
HKU\DEFAULT\Software\Classes\Local Settings\Software\Microsoft\Windows\CurrentVersion\Explorer\Background\Wallpapers\Background\img0.jpg
HKUS-1-5-21-2013161036-436689623-2610530792-1001\Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist\{CEBFF5CD-ACE2-4F4F-9178-9926F41749EA}\CountP:VHrrefZnyjner@rfgbcfnczyr3.rkr: 00 00 00 01 00 00 00 00 00 00 00 00 00
HKUS-1-5-21-2013161036-436689623-2610530792-1001\Software\Microsoft\Windows\CurrentVersion\Explorer\Wallpapers\Background\img0.jpg
HKUS-1-5-21-2013161036-436689623-2610530792-1001\Software\Microsoft\Windows\CurrentVersion\Notifications\Settings\Microsoft.Explorer.Notification.[8B5EAC42-9897-2086-E282-51D34982E322]\LastNotificationAddedTime: 01 82 D7 A8 A3 AE D8 01
HKUS-1-5-21-2013161036-436689623-2610530792-1001\Software\Microsoft\Windows\CurrentVersion\Notifications\Backup\Microsoft.Explorer.Notification.[8B5EAC42-9897-2086-E282-51D34982E322]\Setting: "s:banner,s:toast,s:audio,c:toast,c:ring"
HKUS-1-5-21-2013161036-436689623-2610530792-1001\Software\Microsoft\Windows\CurrentVersion\Notifications\Backup\Microsoft.Explorer.Notification.[8B5EAC42-9897-2086-E282-51D34982E322]\AppId: desktop
HKUS-1-5-21-2013161036-436689623-2610530792-1001\Software\Microsoft\Windows\CurrentVersion\Notifications\Backup\Microsoft.Explorer.Notification.[8B5EAC42-9897-2086-E282-51D34982E322]\Uuid: "NonInvasivePackage"
HKUS-1-5-21-2013161036-436689623-2610530792-1001\Software\Microsoft\Windows\CurrentVersion\CompatFlags\CompatibilityAssistant\Store\C:\Users\malware\Desktop\sample3.exe: 53 41 43 50 01 00 00 00 00 00 00 00 07 00 00 00 28 00 0 0
HKUS-1-5-21-2013161036-436689623-2610530792-1001\Software\Classes\Local Settings\Software\Microsoft\Windows\CurrentVersion\TrayNotify\ToastAppIdentities\Microsoft.Explorer.Notification.[8B5EAC42-9897-2086-E282-51D34982E322]\Icon: "Connection Manager"
HKUS-1-5-21-2013161036-436689623-2610530792-1001\Classes\Local Settings\Software\Microsoft\Windows\CurrentVersion\TrayNotify\ToastAppIdentities\Microsoft.Explorer.Notification.[8B5EAC42-9897-2086-E282-51D34982E322]\Icon: "Connection Manager"
HKUS-1-5-21-2013161036-436689623-2610530792-1001\Classes\Local Settings\Software\Microsoft\Windows\CurrentVersion\TrayNotify\ToastAppIdentities\Microsoft.Explorer.Notification.[8B5EAC42-9897-2086-E282-51D34982E322]\Icon: "Connection Manager"
HKUS-1-5-18\Software\Classes\Local Settings\Software\Microsoft\Windows\CurrentVersion\TrayNotify\ToastAppIdentities\Microsoft.Explorer.Notification.[8B5EAC42-9897-2086-E282-51D34982E322]\Icon: "Contacts"
HKUS-1-5-18\Software\Classes\Local Settings\Software\Microsoft\Windows\CurrentVersion\TrayNotify\ToastAppIdentities\Microsoft.Explorer.Notification.[8B5EAC42-9897-2086-E282-51D34982E322]\Icon: "Contacts"
HKUS-1-5-18\Software\Classes\Local Settings\Software\Microsoft\Windows\CurrentVersion\TrayNotify\ToastAppIdentities\Microsoft.Explorer.Notification.[8B5EAC42-9897-2086-E282-51D34982E322]\Icon: "Saved Pictures"
HKUS-1-5-18\Software\Classes\Local Settings\Software\Microsoft\Windows\CurrentVersion\TrayNotify\ToastAppIdentities\Microsoft.Explorer.Notification.[8B5EAC42-9897-2086-E282-51D34982E322]\Icon: "Camera Roll"
```

Sample3 has two separate instance. The first one is mainly responsible for registry changes, and after it's done, another sample3.exe starts under dllhost.

Process Hacker.exe (1344)	Process Hacker	C:\Users\malware...	wj3Z	2022\malware /-z
sample3.exe (2808)		C:\Users\malware...		2022\malware 7-2
DllHost.exe (1948)	COM Surrogate	C:\Windows\sys...		Microsoft Corporat... NT AUTHORITY\...
DllHost.exe (1376)	COM Surrogate	C:\Windows\sys...		Microsoft Corporat... 2022\malware 7-2\...
sample3.exe (2768)		C:\Users\malware...		2022\malware 7-2\... "C:\...
cmd.exe (3156)	Windows Comma...	C:\Windows\sys...		Microsoft Corporat... 2022\malware 7-2\... "C:\...
DllHost.exe (3244)	COM Surrogate	C:\Windows\sys...		Microsoft Corporat... 2022\malware 7-2\...
DllHost.exe (2128)	COM Surrogate	C:\Windows\sys...		Microsoft Corporat... 2022\malware 7-2\...

This second sample3 is responsible for the main file encryptions. It first creates the .ico file, and then iterates through the directory and encrypts every file. (Following SS filtered to include operations related to 88c86038)

Time of Day	Process Name	PID	Operation	Path	Result	Detail
11:21:01.080295 PM	sample3.exe	2768	CreateFile	C:\Users\malware\AppData\Local\88c86038.ico	SUCCESS	Desired Access: G...
11:21:01.080414 PM	sample3.exe	2768	WriteFile	C:\Users\malware\AppData\Local\88c86038.ico	SUCCESS	Offset: 0, Length: 3...
11:21:01.080471 PM	sample3.exe	2768	CloseFile	C:\Users\malware\AppData\Local\88c86038.ico	SUCCESS	
11:21:01.080539 PM	sample3.exe	2768	RegOpenKey	HKCU\Software\Classes\88c86038	NAME NOT FOUND	Desired Access: M...
11:21:01.080545 PM	sample3.exe	2768	RegCreateKey	HKCR\88c86038	SUCCESS	Desired Access: M...
11:21:01.080765 PM	sample3.exe	2768	RegQueryKey	HKCR\88c86038	SUCCESS	Query Name
11:21:01.080772 PM	sample3.exe	2768	RegOpenKey	HKCU\Software\Classes\88c86038	NAME NOT FOUND	Desired Access: M...
11:21:01.080775 PM	sample3.exe	2768	RegSetValue	HKCR\88c86038 (Default)	SUCCESS	Type: REG_SZ, Le...
11:21:01.080902 PM	sample3.exe	2768	RegCloseKey	HKCR\88c86038	SUCCESS	
11:21:01.080903 PM	sample3.exe	2768	RegOpenKey	HKCU\Software\Classes\88c86038-DefaultIcon	NAME NOT FOUND	Desired Access: M...
11:21:01.080917 PM	sample3.exe	2768	RegCreateKey	HKCR\88c86038-DefaultIcon	SUCCESS	Desired Access: M...
11:21:01.080922 PM	sample3.exe	2768	RegOpenKey	HKCR\88c86038-DefaultIcon	SUCCESS	Desired Access: M...
11:21:01.081086 PM	sample3.exe	2768	RegCloseKey	HKCR\88c86038-DefaultIcon	SUCCESS	Desired Access: M...
11:21:01.081242 PM	sample3.exe	2768	RegOpenKey	HKCR\88c86038	SUCCESS	
11:21:01.081248 PM	sample3.exe	2768	RegQueryKey	HKCR\88c86038-DefaultIcon	SUCCESS	Query Name
11:21:01.081257 PM	sample3.exe	2768	RegOpenKey	HKCU\Software\Classes\88c86038-DefaultIcon	NAME NOT FOUND	Desired Access: M...
11:21:01.081261 PM	sample3.exe	2768	RegSetValue	HKCR\88c86038 (Default)	SUCCESS	Type: REG_SZ, Le...
11:21:04.073609 PM	sample3.exe	2768	CreateFile	C:\README 88c86038.TXT	SUCCESS	Desired Access: G...
11:21:04.073743 PM	sample3.exe	2768	WriteFile	C:\README 88c86038.TXT	SUCCESS	Offset: 0, Length: 1...
11:21:04.073810 PM	sample3.exe	2768	CloseFile	C:\README 88c86038.TXT	SUCCESS	
11:21:05.033012 PM	sample3.exe	2768	CreateFile	C:\python27\README 88c86038.TXT	SUCCESS	Desired Access: G...
11:21:05.033184 PM	sample3.exe	2768	WriteFile	C:\python27\README 88c86038.TXT	SUCCESS	Offset: 0, Length: 1...
11:21:05.033201 PM	sample3.exe	2768	CloseFile	C:\python27\README 88c86038.TXT	SUCCESS	
11:21:05.500336 PM	sample3.exe	2768	CreateFile	C:\python27\Doc\README 88c86038.TXT	SUCCESS	Desired Access: G...
11:21:05.500719 PM	sample3.exe	2768	WriteFile	C:\python27\Doc\README 88c86038.TXT	SUCCESS	Offset: 0, Length: 1...
11:21:05.500921 PM	sample3.exe	2768	CloseFile	C:\python27\Doc\README 88c86038.TXT	SUCCESS	
11:21:05.500957 PM	sample3.exe	2768	CreateFile	C:\python27\Doc\python272.chm 88c86038	SUCCESS	
11:21:05.507504 PM	sample3.exe	2768	ReadFile	C:\python27\Doc\python272.chm 88c86038	SUCCESS	Desired Access: G...
11:21:05.512097 PM	sample3.exe	2768	ReadFile	C:\python27\Doc\python272.chm 88c86038	SUCCESS	Offset: 0, Length: 5...
11:21:05.512158 PM	sample3.exe	2768	ReadFile	C:\python27\Doc\python272.chm 88c86038	SUCCESS	Offset: 0, Length: 5...
11:21:05.5184018 PM	sample3.exe	2768	WriteFile	C:\python27\Doc\python272.chm 88c86038	SUCCESS	Offset: 524,288, Le...
11:21:05.519546 PM	sample3.exe	2768	ReadFile	C:\python27\Doc\python272.chm 88c86038	SUCCESS	Offset: 524,288, Le...
11:21:05.525405 PM	sample3.exe	2768	WriteFile	C:\python27\Doc\python272.chm 88c86038	SUCCESS	Offset: 1,048,576...
11:21:05.525475 PM	sample3.exe	2768	ReadFile	C:\python27\Doc\python272.chm 88c86038	SUCCESS	Offset: 1,048,576...
11:21:05.529434 PM	sample3.exe	2768	WriteFile	C:\python27\Doc\python272.chm 88c86038	SUCCESS	Offset: 1,572,864...
11:21:05.529626 PM	sample3.exe	2768	ReadFile	C:\python27\Doc\python272.chm 88c86038	SUCCESS	Offset: 1,572,864...
11:21:05.531213 PM	sample3.exe	2768	WriteFile	C:\python27\Doc\python272.chm 88c86038	SUCCESS	Offset: 1,572,864...
11:21:05.533341 PM	sample3.exe	2768	ReadFile	C:\python27\Doc\python272.chm 88c86038	SUCCESS	Offset: 2,097,152...
11:21:05.535426 PM	sample3.exe	2768	WriteFile	C:\python27\Doc\python272.chm 88c86038	SUCCESS	Offset: 2,097,152...
11:21:05.535453 PM	sample3.exe	2768	ReadFile	C:\python27\Doc\python272.chm 88c86038	SUCCESS	Offset: 2,621,440...
11:21:05.540035 PM	sample3.exe	2768	WriteFile	C:\python27\Doc\python272.chm 88c86038	SUCCESS	Offset: 2,621,440...
11:21:05.540274 PM	sample3.exe	2768	ReadFile	C:\python27\Doc\python272.chm 88c86038	SUCCESS	Offset: 3,145,728...
11:21:05.543308 PM	sample3.exe	2768	WriteFile	C:\python27\Doc\python272.chm 88c86038	SUCCESS	Offset: 3,145,728...
11:21:05.5434913 PM	sample3.exe	2768	ReadFile	C:\python27\Doc\python272.chm 88c86038	SUCCESS	Offset: 3,670,016...
11:21:05.547915 PM	sample3.exe	2768	WriteFile	C:\python27\Doc\python272.chm 88c86038	SUCCESS	Offset: 3,670,016...
11:21:05.548134 PM	sample3.exe	2768	ReadFile	C:\python27\Doc\python272.chm 88c86038	SUCCESS	Offset: 4,194,304...
11:21:05.551326 PM	sample3.exe	2768	WriteFile	C:\python27\Doc\python272.chm 88c86038	SUCCESS	Offset: 4,194,304...
11:21:05.551524 PM	sample3.exe	2768	ReadFile	C:\python27\Doc\python272.chm 88c86038	SUCCESS	Offset: 4,718,592...
11:21:05.554426 PM	sample3.exe	2768	WriteFile	C:\python27\Doc\python272.chm 88c86038	SUCCESS	Offset: 4,718,592...
11:21:05.554650 PM	sample3.exe	2768	ReadFile	C:\python27\Doc\python272.chm 88c86038	SUCCESS	Offset: 5,242,880...
11:21:05.557644 PM	sample3.exe	2768	WriteFile	C:\python27\Doc\python272.chm 88c86038	SUCCESS	Offset: 5,242,880...
11:21:05.557894 PM	sample3.exe	2768	ReadFile	C:\python27\Doc\python272.chm 88c86038	SUCCESS	Offset: 5,767,168...
11:21:05.558331 PM	sample3.exe	2768	WriteFile	C:\python27\Doc\python272.chm 88c86038	SUCCESS	Offset: 5,767,168...

It also created multiple cmd and powershell (powershell only in win 10) instances, which further created conhost.exe instances (Notice the parent id detail in the first “Process start” operation of each,

which confirms that it was started by the second (under dllhost) sample3 (id 2144 in this run) (for powershell and cmd) and started by powershell and cmd (for conhost.exe)).

Time ...	Process Name	PID	Operation	Path	Result	Detail
12:46:...	powershell.exe	3568	Process Start		SUCCESS	Parent PID: 2144, ...
12:46:...	powershell.exe	3568	Thread Create		SUCCESS	Thread ID: 3152
12:46:...	powershell.exe	3568	Load Image	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	SUCCESS	Image Base: 0x7f7...
12:46:...	powershell.exe	3568	Load Image	C:\Windows\System32\ntdll.dll	SUCCESS	Image Base: 0x7f7a...
12:46:...	powershell.exe	3568	CreateFile	C:\Windows\Prefetch\POWERSHELL.EXE-920BBA2A.pf	SUCCESS	Desired Access: G...
12:46:...	powershell.exe	3568	QueryStandardInfor...	C:\Windows\Prefetch\POWERSHELL.EXE-920BBA2A.pf	SUCCESS	AllocationSize: 40...
12:46:...	powershell.exe	3568	ReadFile	C:\Windows\Prefetch\POWERSHELL.EXE-920BBA2A.pf	SUCCESS	Offset: 0, Length: 4...
12:46:...	powershell.exe	3568	ReadFile	C:\Windows\Prefetch\POWERSHELL.EXE-920BBA2A.pf	SUCCESS	Offset: 0, Length: 4...
12:46:...	powershell.exe	3568	CloseFile	C:\Windows\Prefetch\POWERSHELL.EXE-920BBA2A.pf	SUCCESS	
12:46:...	powershell.exe	3568	RegOpenKey	HKLM\System\CurrentControlSet\Control\Session Manager	REPARSE	Desired Access: Q...
12:46:...	powershell.exe	3568	RegOpenKey	HKLM\System\CurrentControlSet\Control\Session Manager	SUCCESS	Desired Access: Q...
12:46:...	powershell.exe	3568	RegQueryValue	HKLM\System\CurrentControlSet\Control\Session Manager\RaiseExceptionOnPossi...	NAME NOT FOUND	Length: 80
12:46:...	powershell.exe	3568	RegCloseKey	HKLM\System\CurrentControlSet\Control\Session Manager	SUCCESS	
12:46:...	powershell.exe	3568	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\Segment Heap	REPARSE	Desired Access: Q...
12:46:...	powershell.exe	3568	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\Segment Heap	NAME NOT FOUND	Desired Access: Q...
12:46:...	powershell.exe	3568	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager	REPARSE	Desired Access: Q...
12:46:...	powershell.exe	3568	RegOpenKey	HKLM\System\CurrentControlSet\Control\Session Manager	SUCCESS	Desired Access: Q...
12:46:...	powershell.exe	3568	RegQueryValue	HKLM\System\CurrentControlSet\Control\Session Manager\ResourcePolicies	NAME NOT FOUND	Length: 24
12:46:...	powershell.exe	3568	RegCloseKey	HKLM\System\CurrentControlSet\Control\Session Manager	SUCCESS	
12:46:...	powershell.exe	3568	CreateFile	C:\Windows\System32	SUCCESS	Desired Access: E...
12:46:...	powershell.exe	3568	Load Image	C:\Windows\System32\kernel32.dll	SUCCESS	Image Base: 0x7ffa...
12:46:...	powershell.exe	3568	Load Image	C:\Windows\System32\KernelBase.dll	SUCCESS	Image Base: 0x7ffa...
12:46:...	powershell.exe	3568	CreateFile	C:\Windows\System32\conhost.exe	SUCCESS	Desired Access: E...
12:46:...	powershell.exe	3568	CreateFileMapping	C:\Windows\System32\conhost.exe	FILE LOCKED WI...	SyncType: SyncTy...
12:46:...	powershell.exe	3568	CreateFileMapping	C:\Windows\System32\conhost.exe	SUCCESS	SyncType: SyncTy...
12:46:...	powershell.exe	3568	RegOpenKey	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution O...	NAME NOT FOUND	Desired Access: Q...
12:46:...	powershell.exe	3568	QueryNameInformat...	C:\Windows\System32\conhost.exe	SUCCESS	Name: 'Windows'...
12:46:...	powershell.exe	3568	RegOpenKey	HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\S-1-5-21-20131...	SUCCESS	Desired Access: All...
12:46:...	powershell.exe	3568	RegQueryValue	HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\S-1-5-21-20131...	NAME NOT FOUND	Length: 40
12:46:...	powershell.exe	3568	RegCloseKey	HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\S-1-5-21-20131...	SUCCESS	
12:46:...	powershell.exe	3568	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\BAM	REPARSE	Desired Access: Q...
12:46:...	powershell.exe	3568	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\BAM	NAME NOT FOUND	Desired Access: Q...
12:46:...	powershell.exe	3568	Process Create	C:\Windows\System32\conhost.exe	SUCCESS	PID: 7928, Comma...
12:46:...	powershell.exe	3568	CloseFile	C:\Windows\System32\conhost.exe	SUCCESS	
12:46:...	powershell.exe	3568	RegOpenKey	HKLM\System\CurrentControlSet\Control\SafeBoot\Option	REPARSE	Desired Access: Q...
12:46:...	powershell.exe	3568	RegOpenKey	HKLM\System\CurrentControlSet\Control\SafeBoot\Option	NAME NOT FOUND	Desired Access: Q...
12:46:...	powershell.exe	3568	RegOpenKey	HKLM\System\CurrentControlSet\Control\Srp\GP\DLL	REPARSE	Desired Access: R...
12:46:...	powershell.exe	3568	RegOpenKey	HKLM\System\CurrentControlSet\Control\Srp\GP\DLL	NAME NOT FOUND	Desired Access: R...
12:46:...	powershell.exe	3568	RegOpenKey	HKLM\Software\Policies\Microsoft\Windows\Safer\CodeIdentifiers	SUCCESS	Desired Access: Q...
12:46:...	powershell.exe	3568	RegQueryValue	HKLM\SOFTWARE\Policies\Microsoft\Windows\Safer\CodeIdentifiers\Transparent...	NAME NOT FOUND	Length: 80
12:46:...	powershell.exe	3568	RegCloseKey	HKLM\SOFTWARE\Policies\Microsoft\Windows\Safer\CodeIdentifiers	SUCCESS	
12:46:...	powershell.exe	3568	RegOpenKey	HKCU\Software\Policies\Microsoft\Windows\Safer\CodeIdentifiers	NAME NOT FOUND	Desired Access: Q...
12:46:...	powershell.exe	3568	RegOpenKey	HKLM\System\CurrentControlSet\Control\FileSystem\	REPARSE	Desired Access: R...
12:46:...	powershell.exe	3568	RegOpenKey	HKLM\System\CurrentControlSet\Control\FileSystem\	SUCCESS	Desired Access: R...
12:46:...	powershell.exe	3568	RegQueryValue	HKLM\System\CurrentControlSet\Control\FileSystem\LongPathsEnabled	SUCCESS	Type: REG_DWO...
12:46:...	powershell.exe	3568	RegCloseKey	HKLM\System\CurrentControlSet\Control\FileSystem\	SUCCESS	
12:46:...	powershell.exe	3568	RegOpenKey	HKLM\System\CurrentControlSet\Control\FileSystem\	REPARSE	Desired Access: R...
12:46:...	powershell.exe	3568	RegOpenKey	HKLM\System\CurrentControlSet\Control\FileSystem\	SUCCESS	Desired Access: R...
12:46:...	powershell.exe	3568	RegQueryValue	HKLM\System\CurrentControlSet\Control\FileSystem\LPGO	NAME NOT FOUND	Length: 20
12:46:...	powershell.exe	3568	RegCloseKey	HKLM\System\CurrentControlSet\Control\FileSystem\	SUCCESS	
12:46:...	powershell.exe	3568	Load Image	C:\Windows\System32\msvcrt.dll	SUCCESS	Image Base: 0x7ffa...
12:46:...	powershell.exe	3568	Thread Create		SUCCESS	Thread ID: 5288
12:46:...	powershell.exe	3568	Load Image	C:\Windows\System32\oleaut32.dll	SUCCESS	Image Base: 0x7ffa...
12:46:...	Conhost.exe	7928	Process Start		SUCCESS	Parent PID: 3568, ...
12:46:...	Conhost.exe	7928	Thread Create		SUCCESS	Thread ID: 8484
12:46:...	Conhost.exe	7928	Load Image	C:\Windows\System32\conhost.exe	SUCCESS	Image Base: 0x7f7...
12:46:...	Conhost.exe	7928	Load Image	C:\Windows\System32\ntdll.dll	SUCCESS	Image Base: 0x7f7a...
12:46:...	Conhost.exe	7928	CreateFile	C:\Windows\Prefetch\CONHOST.EXE-1F3E9D7E.pf	SUCCESS	Desired Access: G...
12:46:...	Conhost.exe	7928	QueryStandardInfor...	C:\Windows\Prefetch\CONHOST.EXE-1F3E9D7E.pf	SUCCESS	AllocationSize: 12...
12:46:...	Conhost.exe	7928	ReadFile	C:\Windows\Prefetch\CONHOST.EXE-1F3E9D7E.pf	SUCCESS	Offset: 0, Length: 1...
12:46:...	Conhost.exe	7928	CloseFile	C:\Windows\Prefetch\CONHOST.EXE-1F3E9D7E.pf	SUCCESS	
12:46:...	Conhost.exe	7928	RegOpenKey	HKLM\System\CurrentControlSet\Control\Session Manager	REPARSE	Desired Access: Q...
12:46:...	Conhost.exe	7928	RegOpenKey	HKLM\System\CurrentControlSet\Control\Session Manager	SUCCESS	Desired Access: Q...
12:46:...	Conhost.exe	7928	RegQueryValue	HKLM\System\CurrentControlSet\Control\Session Manager\RaiseExceptionOnPossi...	NAME NOT FOUND	Length: 80
12:46:...	Conhost.exe	7928	RegCloseKey	HKLM\System\CurrentControlSet\Control\Session Manager	SUCCESS	
12:46:...	Conhost.exe	7928	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\Segment Heap	REPARSE	Desired Access: Q...
12:46:...	Conhost.exe	7928	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\Segment Heap	NAME NOT FOUND	Desired Access: Q...
12:46:...	Conhost.exe	7928	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager	REPARSE	Desired Access: Q...
12:46:...	Conhost.exe	7928	RegOpenKey	HKLM\System\CurrentControlSet\Control\Session Manager	SUCCESS	Desired Access: Q...
12:46:...	Conhost.exe	7928	RegQueryValue	HKLM\System\CurrentControlSet\Control\Session Manager\ResourcePolicies	NAME NOT FOUND	Length: 24
12:46:...	Conhost.exe	7928	RegCloseKey	HKLM\System\CurrentControlSet\Control\Session Manager	SUCCESS	
12:46:...	Conhost.exe	7928	CreateFile	C:\Windows	SUCCESS	Desired Access: E...
12:46:...	Conhost.exe	7928	Load Image	C:\Windows\System32\kernel32.dll	SUCCESS	Image Base: 0x7ffa...
12:46:...	Conhost.exe	7928	Load Image	C:\Windows\System32\KernelBase.dll	SUCCESS	Image Base: 0x7ffa...
12:46:...	Conhost.exe	7928	RegOpenKey	HKLM\System\CurrentControlSet\Control\SafeBoot\Option	REPARSE	Desired Access: Q...
12:46:...	Conhost.exe	7928	RegOpenKey	HKLM\System\CurrentControlSet\Control\SafeBoot\Option	NAME NOT FOUND	Desired Access: Q...
12:46:...	Conhost.exe	7928	RegOpenKey	HKLM\System\CurrentControlSet\Control\Srp\GP\DLL	REPARSE	Desired Access: R...
12:46:...	Conhost.exe	7928	RegOpenKey	HKLM\System\CurrentControlSet\Control\Srp\GP\DLL	NAME NOT FOUND	Desired Access: R...
12:46:...	Conhost.exe	7928	RegOpenKey	HKLM\Software\Policies\Microsoft\Windows\Safer\CodeIdentifiers	SUCCESS	Desired Access: Q...
12:46:...	Conhost.exe	7928	RegQueryValue	HKLM\SOFTWARE\Policies\Microsoft\Windows\Safer\CodeIdentifiers\Transparent...	NAME NOT FOUND	Length: 80
12:46:...	Conhost.exe	7928	RegCloseKey	HKLM\SOFTWARE\Policies\Microsoft\Windows\Safer\CodeIdentifiers	SUCCESS	
12:46:...	Conhost.exe	7928	RegOpenKey	HKCU\Software\Policies\Microsoft\Windows\Safer\CodeIdentifiers	NAME NOT FOUND	Desired Access: Q...
12:46:...	Conhost.exe	7928	RegOpenKey	HKLM\System\CurrentControlSet\Control\FileSystem\	REPARSE	Desired Access: R...
12:46:...	Conhost.exe	7928	RegOpenKey	HKLM\System\CurrentControlSet\Control\FileSystem\	SUCCESS	Desired Access: R...
12:46:...	Conhost.exe	7928	RegQueryValue	HKLM\System\CurrentControlSet\Control\FileSystem\LongPathsEnabled	SUCCESS	Type: REG_DWO...
12:46:...	Conhost.exe	7928	RegCloseKey	HKLM\System\CurrentControlSet\Control\FileSystem\	SUCCESS	
12:46:...	Conhost.exe	7928	RegOpenKey	HKLM\System\CurrentControlSet\Control\FileSystem\	REPARSE	Desired Access: R...
12:46:...	Conhost.exe	7928	RegOpenKey	HKLM\System\CurrentControlSet\Control\FileSystem\	SUCCESS	Desired Access: R...
12:46:...	Conhost.exe	7928	RegQueryValue	HKLM\System\CurrentControlSet\Control\FileSystem\LPGO	NAME NOT FOUND	Length: 20
12:46:...	Conhost.exe	7928	RegCloseKey	HKLM\System\CurrentControlSet\Control\FileSystem\	SUCCESS	
12:46:...	Conhost.exe	7928	Load Image	C:\Windows\System32\msvcrt.dll	SUCCESS	Image Base: 0x7ffa...
12:46:...	Conhost.exe	7928	Load Image	C:\Windows\System32\userbase.dll	SUCCESS	Image Base: 0x7ffa...
12:46:...	Conhost.exe	7928	Thread Create		SUCCESS	Thread ID: 6688
12:46:...	Conhost.exe	7928	Load Image	C:\Windows\System32\SHCore.dll	SUCCESS	Image Base: 0x7ffa...
12:46:...	Conhost.exe	7928	Load Image	C:\Windows\System32\msvcrt.dll	SUCCESS	Image Base: 0x7ffa...
12:46:...	Conhost.exe	7928	Load Image	C:\Windows\System32\combase.dll	SUCCESS	Image Base: 0x7ffa...
12:46:...	Conhost.exe	7928	Load Image	C:\Windows\System32\iprt4.dll	SUCCESS	Image Base: 0x7ffa...
12:46:...	Conhost.exe	7928	Load Image	C:\Windows\System32\advapi32.dll	SUCCESS	Image Base: 0x7ffa...
12:46:...	Conhost.exe	7928	Load Image	C:\Windows\System32\sechost.dll	SUCCESS	Image Base: 0x7ffa...
12:46:...	Conhost.exe	7928	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager	REPARSE	Desired Access: Q...

Time ...	Process Name	PID	Operation	Path	Result	Detail
12:49:...	cmd.exe	4656	Process Start		SUCCESS	Parent PID: 2144, Comm...
12:49:...	cmd.exe	4656	Thread Create		SUCCESS	Thread ID: 3876
12:49:...	cmd.exe	4656	Load Image	C:\Windows\SysWOW64\cmd.exe	SUCCESS	Image Base: 0x00000, I...
12:49:...	cmd.exe	4656	Load Image	C:\Windows\System32\ntdll.dll	SUCCESS	Image Base: 0x7faa619...
12:49:...	cmd.exe	4656	Load Image	C:\Windows\SysWOW64\ntdll.dll	SUCCESS	Image Base: 0x778e000...
12:49:...	cmd.exe	4656	CreateFile	C:\Windows\Prefetch\CMD EXE-AC113AA8.pf	NAME NOT FOUND	Desired Access: Generic...
12:49:...	cmd.exe	4656	RegOpenKey	HKLM\System\CurrentControlSet\Control\Session Manager	REPARSE	Desired Access: Query V...
12:49:...	cmd.exe	4656	RegOpenKey	HKLM\System\CurrentControlSet\Control\Session Manager	SUCCESS	Desired Access: Query V...
12:49:...	cmd.exe	4656	RegQueryValue	HKLM\System\CurrentControlSet\Control\Session Manager\RaiseExceptionOnPossi...	NAME NOT FOUND	Length: 80
12:49:...	cmd.exe	4656	RegCloseKey	HKLM\System\CurrentControlSet\Control\Session Manager	SUCCESS	
12:49:...	cmd.exe	4656	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\Segment Heap	REPARSE	Desired Access: Query V...
12:49:...	cmd.exe	4656	RegOpenKey	HKLM\System\CurrentControlSet\Control\Session Manager\Segment Heap	NAME NOT FOUND	Desired Access: Query V...
12:49:...	cmd.exe	4656	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager	REPARSE	Desired Access: Query V...
12:49:...	cmd.exe	4656	RegOpenKey	HKLM\System\CurrentControlSet\Control\Session Manager	SUCCESS	Desired Access: Query V...
12:49:...	cmd.exe	4656	RegQueryValue	HKLM\System\CurrentControlSet\Control\Session Manager\ResourcePolicies	NAME NOT FOUND	Length: 24
12:49:...	cmd.exe	4656	RegCloseKey	HKLM\System\CurrentControlSet\Control\Session Manager	SUCCESS	
12:49:...	cmd.exe	4656	CreateFile	C:\Windows	SUCCESS	Desired Access: Execute...
12:49:...	cmd.exe	4656	Load Image	C:\Windows\System32\wow64.dll	SUCCESS	Image Base: 0x7faa4eb...
12:49:...	cmd.exe	4656	ReadFile	C:\Windows\System32\wow64.dll	SUCCESS	Offset: 242,176, Length: ...
12:49:...	cmd.exe	4656	Load Image	C:\Windows\System32\wow64win.dll	SUCCESS	Image Base: 0x7faa4c0...
12:49:...	cmd.exe	4656	ReadFile	C:\Windows\System32\wow64win.dll	SUCCESS	Offset: 449,024, Length: ...
12:49:...	cmd.exe	4656	ReadFile	C:\Windows\System32\wow64win.dll	SUCCESS	Offset: 323,584, Length: ...
12:49:...	cmd.exe	4656	CreateFile	C:\Windows\System32\wow64log.dll	NAME NOT FOUND	Desired Access: Read At...
12:49:...	cmd.exe	4656	ReadFile	C:\Windows\System32\wow64.dll	SUCCESS	Offset: 337,920, Length: ...
12:49:...	cmd.exe	4656	CreateFile	C:\Windows	SUCCESS	Desired Access: Read At...
12:49:...	cmd.exe	4656	QueryNameInformat...	C:\Windows	SUCCESS	Name: \Windows
12:49:...	cmd.exe	4656	CloseFile	C:\Windows	SUCCESS	
12:49:...	cmd.exe	4656	RegOpenKey	HKLM\Software\Microsoft\Wow64\86	SUCCESS	Desired Access: Read
12:49:...	cmd.exe	4656	RegQueryValue	HKLM\SOFTWARE\Microsoft\Wow64\86\cmd.exe	NAME NOT FOUND	Length: 520
12:49:...	cmd.exe	4656	RegQueryValue	HKLM\SOFTWARE\Microsoft\Wow64\86(Default)	SUCCESS	Type: REG_SZ, Length: ...
12:49:...	cmd.exe	4656	RegCloseKey	HKLM\SOFTWARE\Microsoft\Wow64\86	SUCCESS	
12:49:...	cmd.exe	4656	Load Image	C:\Windows\System32\wow64cpu.dll	SUCCESS	Image Base: 0x778d000...
12:49:...	cmd.exe	4656	ReadFile	C:\Windows\System32\wow64cpu.dll	SUCCESS	Offset: 6,144, Length: 3...
12:49:...	cmd.exe	4656	ReadFile	C:\Windows\System32\wow64cpu.dll	SUCCESS	Offset: 9,216, Length: 51...
12:49:...	cmd.exe	4656	RegOpenKey	HKLM\System\CurrentControlSet\Control\Session Manager	REPARSE	Desired Access: Query V...
12:49:...	cmd.exe	4656	RegOpenKey	HKLM\System\CurrentControlSet\Control\Session Manager	SUCCESS	Desired Access: Query V...
12:49:...	cmd.exe	4656	RegSetInfoKey	HKLM\System\CurrentControlSet\Control\Session Manager	SUCCESS	KeySetInformationClass: ...
12:49:...	cmd.exe	4656	RegQueryValue	HKLM\System\CurrentControlSet\Control\Session Manager\RaiseExceptionOnPossi...	NAME NOT FOUND	Length: 80
12:49:...	cmd.exe	4656	RegCloseKey	HKLM\System\CurrentControlSet\Control\Session Manager	SUCCESS	
12:49:...	cmd.exe	4656	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\Segment Heap	REPARSE	Desired Access: Query V...
12:49:...	cmd.exe	4656	RegOpenKey	HKLM\System\CurrentControlSet\Control\Session Manager\Segment Heap	NAME NOT FOUND	Desired Access: Query V...
12:49:...	cmd.exe	4656	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager	REPARSE	Desired Access: Query V...
12:49:...	cmd.exe	4656	RegOpenKey	HKLM\System\CurrentControlSet\Control\Session Manager	SUCCESS	Desired Access: Query V...
12:49:...	cmd.exe	4656	RegSetInfoKey	HKLM\System\CurrentControlSet\Control\Session Manager	SUCCESS	KeySetInformationClass: ...
12:49:...	cmd.exe	4656	RegQueryValue	HKLM\System\CurrentControlSet\Control\Session Manager\ResourcePolicies	NAME NOT FOUND	Length: 24
12:49:...	cmd.exe	4656	RegCloseKey	HKLM\System\CurrentControlSet\Control\Session Manager	SUCCESS	
12:49:...	cmd.exe	4656	ReadFile	C:\Windows\SysWOW64\cmd.exe	SUCCESS	Offset: 181,760, Length: ...
12:49:...	cmd.exe	4656	ReadFile	C:\Windows\SysWOW64\cmd.exe	SUCCESS	Offset: 152,576, Length: ...
12:49:...	cmd.exe	4656	ReadFile	C:\Windows\SysWOW64\cmd.exe	SUCCESS	Offset: 192,000, Length: ...
12:49:...	cmd.exe	4656	ReadFile	C:\Windows\SysWOW64\cmd.exe	SUCCESS	Offset: 182,272, Length: ...
12:49:...	cmd.exe	4656	CreateFile	C:\Windows\SysWOW64	SUCCESS	Desired Access: Events

Time ...	Process Name	PID	Operation	Path	Result	Detail
12:49:...	conhost.exe	8696	Process Start		SUCCESS	Parent PID: 4656, Command line: ??C:\Windows\sys...
12:49:...	conhost.exe	8696	Thread Create		SUCCESS	Thread ID: 1832
12:49:...	conhost.exe	8696	Load Image	C:\Windows\System32\conhost.exe	SUCCESS	Image Base: 0x7f72a310000, Image Size: 0xb000
12:49:...	conhost.exe	8696	Load Image	C:\Windows\System32\ntdll.dll	SUCCESS	Image Base: 0x7faa6190000, Image Size: 0x1f8000
12:49:...	conhost.exe	8696	CreateFile	C:\Windows\Prefetch\CONHOST EXE-1F3E9D7E.pf	SUCCESS	Desired Access: Generic Read, Disposition: Open, Optio...
12:49:...	conhost.exe	8696	QueryStandardInfor...	C:\Windows\Prefetch\CONHOST EXE-1F3E9D7E.pf	SUCCESS	AllocationSize: 12,288, EndOfFile: 10,418, NumberOfLin...
12:49:...	conhost.exe	8696	ReadFile	C:\Windows\Prefetch\CONHOST EXE-1F3E9D7E.pf	SUCCESS	Offset: 0, Length: 10,418, Priority: High
12:49:...	conhost.exe	8696	ReadFile	C:\Windows\Prefetch\CONHOST EXE-1F3E9D7E.pf	SUCCESS	Offset: 0, Length: 10,418, I/O Flags: Non-cached, Pagin...
12:49:...	conhost.exe	8696	CloseFile	C:\Windows\Prefetch\CONHOST EXE-1F3E9D7E.pf	SUCCESS	
12:49:...	conhost.exe	8696	RegOpenKey	HKLM\System\CurrentControlSet\Control\Session Manager	REPARSE	Desired Access: Query Value
12:49:...	conhost.exe	8696	RegOpenKey	HKLM\System\CurrentControlSet\Control\Session Manager	SUCCESS	Desired Access: Query Value
12:49:...	conhost.exe	8696	RegQueryValue	HKLM\System\CurrentControlSet\Control\Session Manager\RaiseExceptionOnPossi...	NAME NOT FOUND	Length: 80
12:49:...	conhost.exe	8696	RegCloseKey	HKLM\System\CurrentControlSet\Control\Session Manager	SUCCESS	
12:49:...	conhost.exe	8696	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\Segment Heap	REPARSE	Desired Access: Query Value
12:49:...	conhost.exe	8696	RegOpenKey	HKLM\System\CurrentControlSet\Control\Session Manager\Segment Heap	NAME NOT FOUND	Desired Access: Query Value
12:49:...	conhost.exe	8696	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager	REPARSE	Desired Access: Query Value, Enumerate Sub Keys
12:49:...	conhost.exe	8696	RegOpenKey	HKLM\System\CurrentControlSet\Control\Session Manager	SUCCESS	Desired Access: Query Value, Enumerate Sub Keys
12:49:...	conhost.exe	8696	RegQueryValue	HKLM\System\CurrentControlSet\Control\Session Manager\ResourcePolicies	NAME NOT FOUND	Length: 24
12:49:...	conhost.exe	8696	RegCloseKey	HKLM\System\CurrentControlSet\Control\Session Manager	SUCCESS	
12:49:...	conhost.exe	8696	ReadFile	C:\Windows\System32\conhost.exe	SUCCESS	Offset: 760,832, Length: 16,384, I/O Flags: Non-cached...
12:49:...	conhost.exe	8696	CreateFile	C:\Windows	SUCCESS	Desired Access: Execute/Traverse, Synchronize, Dispo...
12:49:...	conhost.exe	8696	ReadFile	C:\Windows\System32\conhost.exe	SUCCESS	Offset: 691,200, Length: 16,384, I/O Flags: Non-cached...
12:49:...	conhost.exe	8696	Load Image	C:\Windows\System32\kernel32.dll	SUCCESS	Image Base: 0x7faa4b00000, Image Size: 0xb0000
12:49:...	conhost.exe	8696	Load Image	C:\Windows\System32\KernelBase.dll	SUCCESS	Image Base: 0x7faa3800000, Image Size: 0x26a000
12:49:...	conhost.exe	8696	RegOpenKey	HKLM\System\CurrentControlSet\Control\SafeBoot\Option	REPARSE	Desired Access: Query Value, Set Value
12:49:...	conhost.exe	8696	RegOpenKey	HKLM\System\CurrentControlSet\Control\SafeBoot\Option	NAME NOT FOUND	Desired Access: Query Value, Set Value
12:49:...	conhost.exe	8696	RegOpenKey	HKLM\System\CurrentControlSet\Control\Ssp\GP\DLL	REPARSE	Desired Access: Read
12:49:...	conhost.exe	8696	RegOpenKey	HKLM\System\CurrentControlSet\Control\Ssp\GP\DLL	NAME NOT FOUND	Desired Access: Read
12:49:...	conhost.exe	8696	RegOpenKey	HKLM\Software\Policies\Microsoft\Windows\Safer\CodeIdentifiers	SUCCESS	Desired Access: Query Value
12:49:...	conhost.exe	8696	RegQueryValue	HKLM\SOFTWARE\Policies\Microsoft\Windows\Safer\CodeIdentifiers\Transparent...	NAME NOT FOUND	Length: 80
12:49:...	conhost.exe	8696	RegCloseKey	HKLM\SOFTWARE\Policies\Microsoft\Windows\Safer\CodeIdentifiers	SUCCESS	
12:49:...	conhost.exe	8696	RegOpenKey	HKCU\Software\Policies\Microsoft\Windows\Safer\CodeIdentifiers	NAME NOT FOUND	Desired Access: Query Value
12:49:...	conhost.exe	8696	RegOpenKey	HKLM\System\CurrentControlSet\Control\FileSystem	REPARSE	Desired Access: Read
12:49:...	conhost.exe	8696	RegOpenKey	HKLM\System\CurrentControlSet\Control\FileSystem	SUCCESS	Desired Access: Read
12:49:...	conhost.exe	8696	RegQueryValue	HKLM\System\CurrentControlSet\Control\FileSystem\LongPathsEnabled	SUCCESS	Type: REG_DWORD, Length: 4, Data: 1
12:49:...	conhost.exe	8696	RegCloseKey	HKLM\System\CurrentControlSet\Control\FileSystem	SUCCESS	
12:49:...	conhost.exe	8696	RegOpenKey	HKLM\System\CurrentControlSet\Control\FileSystem	REPARSE	Desired Access: Read
12:49:...	conhost.exe	8696	RegOpenKey	HKLM\System\CurrentControlSet\Control\FileSystem	SUCCESS	Desired Access: Read
12:49:...	conhost.exe	8696	RegQueryValue	HKLM\System\CurrentControlSet\Control\FileSystem\LPGO	NAME NOT FOUND	Length: 20
12:49:...	conhost.exe	8696	RegCloseKey	HKLM\System\CurrentControlSet\Control\FileSystem	SUCCESS	
12:49:...	conhost.exe	8696	Load Image	C:\Windows\System32\msvcrt.dll	SUCCESS	Image Base: 0x7faa4150000, Image Size: 0x9d000
12:49:...	conhost.exe	8696	Load Image	C:\Windows\System32\urlbase.dll	SUCCESS	Image Base: 0x7faa3c60000, Image Size: 0x10000
12:49:...	conhost.exe	8696	Thread Create		SUCCESS	Thread ID: 552
12:49:...	conhost.exe	8696	ReadFile	C:\Windows\System32\conhost.exe	SUCCESS	Offset: 777,216, Length: 6,144, I/O Flags: Non-cached, ...
12:49:...	conhost.exe	8696	Thread Create		SUCCESS	Thread ID: 1788
12:49:...	conhost.exe	8696	Load Image	C:\Windows\System32\SHCore.dll	SUCCESS	Image Base: 0x7faa4100000, Image Size: 0xad000
12:49:...	conhost.exe	8696	Load Image	C:\Windows\System32\msvcrt.dll	SUCCESS	Image Base: 0x7faa4c00000, Image Size: 0x9e000
12:49:...	conhost.exe	8696	Load Image	C:\Windows\System32\combase.dll	SUCCESS	Image Base: 0x7faa4280000, Image Size: 0x354000
12:49:...	conhost.exe	8696	Load Image	C:\Windows\System32\popt4.dll	SUCCESS	Image Base: 0x7faa4c90000, Image Size: 0x126000
12:49:...	conhost.exe	8696	Load Image	C:\Windows\System32\advapi32.dll	SUCCESS	Image Base: 0x7faa5960000, Image Size: 0xae000
12:49:...	conhost.exe	8696	Load Image	C:\Windows\System32\eechost.dll	SUCCESS	Image Base: 0x7faa45e0000, Image Size: 0x3c000
12:49:...	conhost.exe	8696	RegOpenKey	HKLM\System\CurrentControlSet\Control\Session Manager	REPARSE	Desired Access: Query Value, Enumerate Sub Keys
12:49:...	conhost.exe	8696	RegOpenKey	HKLM\System\CurrentControlSet\Control\Session Manager	SUCCESS	Desired Access: Query Value, Enumerate Sub Keys

The loaded DLLs, which could not be determined in PEStudio or Ghidra, were found being loaded here.

11:20:59.9400559 PM	sample3.exe	2808	cpProcess Start		SUCCESS	Parent PID: 2668...
11:20:59.9480575 PM	sample3.exe	2808	cpThread Create		SUCCESS	Thread ID: 3164
11:21:00.0464500 PM	sample3.exe	2808	cpLoad Image	C:\Users\malware\Desktop\sample3.exe	SUCCESS	Image Base: 0x6b0...
11:21:00.0470559 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\ntdll.dll	SUCCESS	Image Base: 0x76e...
11:21:00.0479641 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\kernel32.dll	SUCCESS	Image Base: 0x757...
11:21:00.0482043 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\kernelbase.dll	SUCCESS	Image Base: 0x752...
11:21:00.0491705 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\advapi32.dll	SUCCESS	Image Base: 0x769...
11:21:00.0493122 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\msvrt.dll	SUCCESS	Image Base: 0x756...
11:21:00.0495900 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\sechost.dll	SUCCESS	Image Base: 0x76e...
11:21:00.0501549 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\iprt4.dll	SUCCESS	Image Base: 0x759...
11:21:00.0507211 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\shell32.dll	SUCCESS	Image Base: 0x760...
11:21:00.0509594 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\shlwapi.dll	SUCCESS	Image Base: 0x75b...
11:21:00.0511394 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\gdi32.dll	SUCCESS	Image Base: 0x752...
11:21:00.0513004 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\user32.dll	SUCCESS	Image Base: 0x752...
11:21:00.0515137 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\lpk.dll	SUCCESS	Image Base: 0x752...
11:21:00.0516884 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\usp10.dll	SUCCESS	Image Base: 0x759...
11:21:00.0532149 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\imm32.dll	SUCCESS	Image Base: 0x769...
11:21:00.0534094 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\ole32.dll	SUCCESS	Image Base: 0x759...
11:21:00.1284391 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\ole32.dll	SUCCESS	Image Base: 0x75c...
11:21:00.1288045 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\oleaut32.dll	SUCCESS	Image Base: 0x75c...
11:21:00.1295385 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\mpr.dll	SUCCESS	Image Base: 0x69d...
11:21:00.1302079 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\UPHLPAPI.DLL	SUCCESS	Image Base: 0x77a...
11:21:00.1303667 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\ncsi.dll	SUCCESS	Image Base: 0x756...
11:21:00.1309704 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\winnsi.dll	SUCCESS	Image Base: 0x72d...
11:21:00.1316824 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\netapi32.dll	SUCCESS	Image Base: 0x732... Image Size: 0x12000
11:21:00.1322758 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\ntevents.dll	SUCCESS	Image Base: 0x776...
11:21:00.1328523 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\advapi32.dll	SUCCESS	Image Base: 0x747...
11:21:00.1335679 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\wsxcl.dll	SUCCESS	Image Base: 0x732...
11:21:00.1343095 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\wsxcl.dll	SUCCESS	Image Base: 0x705...
11:21:00.1344625 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\wsx2_32.dll	SUCCESS	Image Base: 0x754...
11:21:00.1345765 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\wininet.dll	SUCCESS	Image Base: 0x74e...
11:21:00.1348451 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\api-ms-win-downlevel-user3211-1-0.dll	SUCCESS	Image Base: 0x74e...
11:21:00.1349600 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\api-ms-win-downlevel-shlwapi11-1-0.dll	SUCCESS	Image Base: 0x74e...
11:21:00.1351058 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\api-ms-win-downlevel-version11-1-0.dll	SUCCESS	Image Base: 0x74e...
11:21:00.1356783 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\version.dll	SUCCESS	Image Base: 0x74e...
11:21:00.1358308 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\api-ms-win-downlevel-normalz11-1-0.dll	SUCCESS	Image Base: 0x74c...
11:21:00.1359582 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\normalz.dll	SUCCESS	Image Base: 0x751...
11:21:00.1361054 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\vertdll.dll	SUCCESS	Image Base: 0x74a...
11:21:00.1362559 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\api-ms-win-downlevel-advapi3211-1-0.dll	SUCCESS	Image Base: 0x74a...
11:21:00.1364983 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\userenv.dll	SUCCESS	Image Base: 0x74e...
11:21:00.1366752 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\profapi.dll	SUCCESS	Image Base: 0x74c...
11:21:00.1374907 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\ntasnapi32.dll	SUCCESS	Image Base: 0x72f...
11:21:00.1389996 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\cryptbase.dll	SUCCESS	Image Base: 0x74a...
11:21:00.1395960 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\udhime.dll	SUCCESS	Image Base: 0x737...
11:21:00.1396370 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\cbcatapi.dll	SUCCESS	Image Base: 0x75a...
11:21:00.1398374 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\comsvcs.dll	SUCCESS	Image Base: 0x69e...
11:21:00.1398803 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\advapi32.dll	SUCCESS	Image Base: 0x769...
11:21:00.1407014 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\api-ms-win-downlevel-advapi3211-1-0.dll	SUCCESS	Image Base: 0x74d...
11:21:00.1428296 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\cryptbase.dll	SUCCESS	Image Base: 0x74a...
11:21:00.1442346 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\api-ms-win-downlevel-advapi3211-1-0.dll	SUCCESS	Image Base: 0x74c...
11:21:00.1566529 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\crypt32.dll	SUCCESS	Image Base: 0x74d...
11:21:00.1567913 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\cryptsp.dll	SUCCESS	Image Base: 0x745...
11:21:00.1654779 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\cryptsp.dll	SUCCESS	Image Base: 0x747...
11:21:00.1655122 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\bcrypt.dll	SUCCESS	Image Base: 0x747...
11:21:00.1655496 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\bcryptprimitives.dll	SUCCESS	Image Base: 0x742...
11:21:00.2101534 PM	sample3.exe	2808	cpThread Create		SUCCESS	Thread ID: 952
11:21:00.2608955 PM	sample3.exe	2808	cpThread Create		SUCCESS	Thread ID: 2008

Multiple cryptographic dlls were also loaded. The earlier ones (advpi32) could have been used for decrypting the original payload. Further dlls could have been used for encrypting the files.

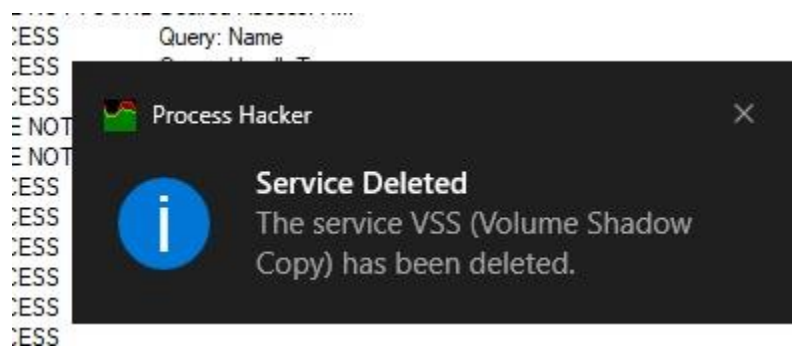
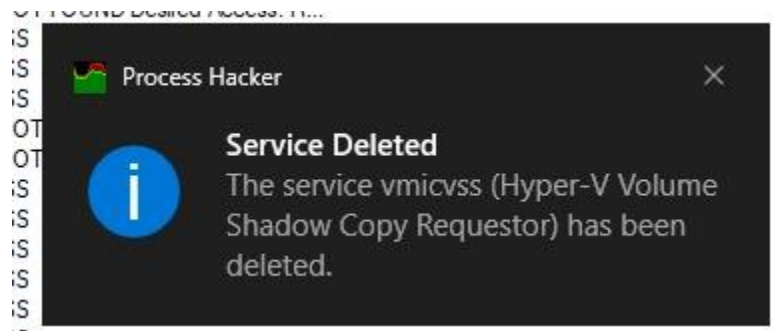
(This screenshot is filtered to get cryptography based dlls)

Time of Day	Process Name	PID	Operation	Path	Result	Detail
11:21:00.0491705 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\advapi32.dll	SUCCESS	Image Base: 0x769...
11:21:00.1362559 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\api-ms-win-downlevel-advapi3211-1-0.dll	SUCCESS	Image Base: 0x74d...
11:21:00.1389996 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\cryptbase.dll	SUCCESS	Image Base: 0x74a...
11:21:00.2002026 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\cryptsp.dll	SUCCESS	Image Base: 0x745...
11:21:00.2598374 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\advapi32.dll	SUCCESS	Image Base: 0x769...
11:21:00.1407014 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\api-ms-win-downlevel-advapi3211-1-0.dll	SUCCESS	Image Base: 0x74d...
11:21:00.1428296 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\cryptbase.dll	SUCCESS	Image Base: 0x74a...
11:21:00.1442346 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\api-ms-win-downlevel-advapi3211-1-0.dll	SUCCESS	Image Base: 0x74c...
11:21:00.1566529 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\crypt32.dll	SUCCESS	Image Base: 0x74d...
11:21:00.1567913 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\cryptsp.dll	SUCCESS	Image Base: 0x745...
11:21:00.1654779 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\cryptsp.dll	SUCCESS	Image Base: 0x747...
11:21:00.1655122 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\bcrypt.dll	SUCCESS	Image Base: 0x747...
11:21:00.1655496 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\bcryptprimitives.dll	SUCCESS	Image Base: 0x742...
11:21:00.1739955 PM	sample3.exe	2808	cpLoad Image	C:\Windows\System32\cryptnet.dll	SUCCESS	Image Base: 0x713...

Most notably, the malware sends a DNS request for securebestapp20.com and temisleys.com

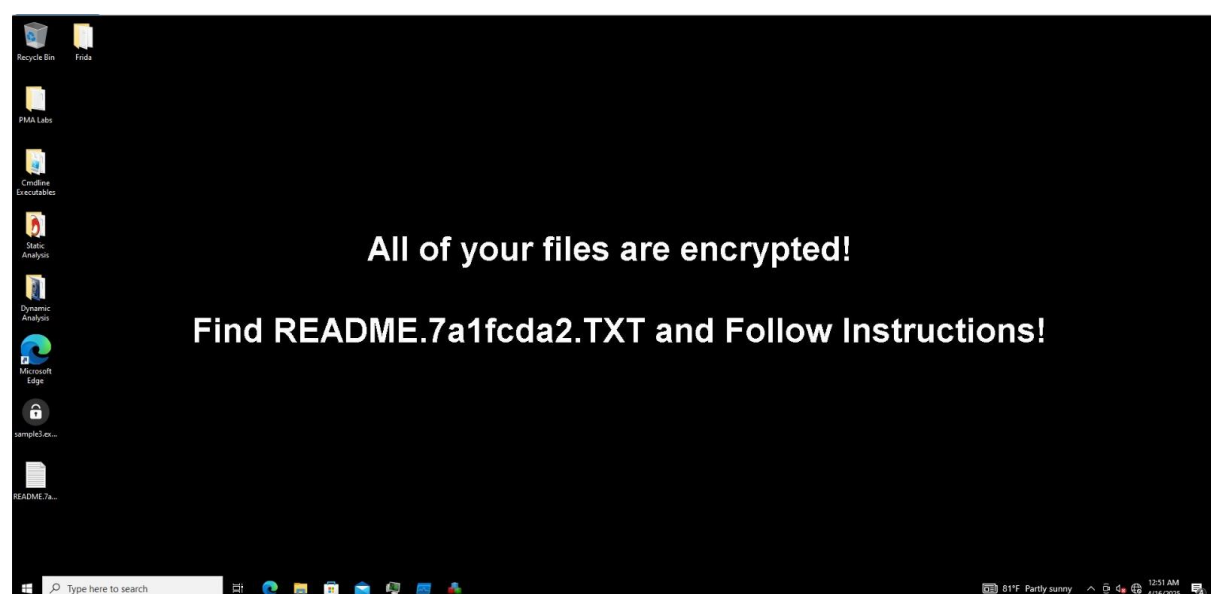
No.	Time	Source	Destination	Protocol	Length	Info
1	0.00000000	192.168.245.133	8.8.8.8	TCP	76	50768 -- 53 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=1343277542 TSecr=0 WS=128
2	2.51518043	192.168.245.129	192.168.245.255	ICMP	84	Name query NS DESKTOP-J0070K1C1<
3	2.73165345	192.168.245.129	192.168.245.133	DNS	81	Standard query 0x8b3f A securebestapp20.com
4	2.71231905	192.168.245.133	192.168.245.129	DNS	97	Standard query response 0x8b3f A securebestapp20.com A 192.168.245.133
5	2.72857148	192.168.245.129	192.168.245.133	TCP	60	8433 -- 443 [ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM=1 WS=128
6	2.73361318	192.168.245.133	192.168.245.129	TCP	60	443 -- 57657 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM=1 WS=128
7	2.73421378	192.168.245.129	192.168.245.133	TCP	62	57657 -- 443 [ACK] Seq=1 Ack=1 Win=262144 Len=0
8	2.74448528	192.168.245.129	192.168.245.133	TLSv1.2	242	Client Hello
9	2.74459184	192.168.245.133	192.168.245.129	TCP	56	443 -- 57657 [ACK] Seq=1 Ack=187 Win=64128 Len=0
10	2.74724912	192.168.245.129	192.168.245.133	TLSv1.2	1354	Server Hello, Certificate, Server Key Exchange, Server Hello Done
11	2.74737391	192.168.245.129	192.168.245.133	TCP	62	57657 -- 443 [ACK] Seq=187 Ack=1299 Win=268608 Len=0
12	2.84332683	127.0.0.1	127.0.0.53	DNS	87	Standard query 0x6b07 A ntp.ubuntu.com OPT
13	2.84334873	127.0.0.1	127.0.0.53	DNS	87	Standard query 0x6b3c AAAA ntp.ubuntu.com OPT
14	2.84337856	127.0.0.53	127.0.0.1	DNS	87	Standard query response 0x632c Server failure A ntp.ubuntu.com OPT
15	2.84339553	127.0.0.53	127.0.0.1	ICMP	115	Destination unreachable (Port unreachable)
16	2.84367991	127.0.0.53	127.0.0.1	DNS	87	Standard query response 0x6b07 Server failure A ntp.ubuntu.com OPT
17	2.84371274	127.0.0.53	127.0.0.1	DNS	87	Standard query response 0xa329 Server failure AAAA ntp.ubuntu.com OPT
18	2.84711917	127.0.0.1	127.0.0.53	ICMP	115	Destination unreachable (Port unreachable)
19	2.84713265	127.0.0.53	127.0.0.1	DNS	115	Destination unreachable (Port unreachable)
20	2.84738332	127.0.0.1	127.0.0.53	ICMP	115	Destination unreachable (Port unreachable)
21	2.84394436	192.168.245.133	8.8.8.8	TCP	76	30956 -- 53 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=1343280385 TSecr=0 WS=128
22	2.93388261	192.168.245.129	192.168.245.133	TCP	60	57658 -- 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
23	2.93394550	192.168.245.133	192.168.245.129	TCP	60	80 -- 57658 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM=1 WS=128
24	2.93416988	192.168.245.129	192.168.245.133	TCP	62	57658 -- 80 [ACK] Seq=1 Ack=1 Win=2182272 Len=0
25	2.93428736	192.168.245.129	192.168.245.133	HTTP	256	GET /msdownload/update/v3/static/trusted/en/disallowedcerts1.cab?69ea938222272056 HTTP/1.1
26	2.93428761	192.168.245.133	192.168.245.129	TCP	56	80 -- 57658 [ACK] Seq=1 Ack=203 Win=64128 Len=0
27	2.96282461	192.168.245.133	192.168.245.129	TCP	266	80 -- 57658 [PSH, ACK] Seq=1 Ack=203 Win=64128 Len=150 [TCP segment of a reassembled PDU]
28	2.96590621	192.168.245.133	192.168.245.129	HTTP	314	HTTP/1.1 200 OK (text/html)
29	2.96565080	192.168.245.129	192.168.245.133	TCP	62	57658 -- 80 [ACK] Seq=203 Ack=410 Win=2181760 Len=0
30	2.96651105	192.168.245.129	192.168.245.133	TCP	62	57658 -- 80 [FIN, ACK] Seq=203 Ack=410 Win=2181760 Len=0
31	2.96653153	192.168.245.133	192.168.245.129	TCP	56	80 -- 57658 [ACK] Seq=410 Ack=284 Win=64128 Len=0
32	2.98115914	192.168.245.129	192.168.245.133	TCP	60	57659 -- 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
49	3.04521527	192.168.245.129	192.168.245.133	TCP	62	57657 -- 443 [ACK] Seq=1031 Ack=1705 Win=261888 Len=0
50	3.04522672	192.168.245.133	192.168.245.129	TLSv1.2	343	Application Data
51	3.04530915	192.168.245.129	192.168.245.133	TCP	62	57657 -- 443 [ACK] Seq=1031 Ack=1992 Win=261632 Len=0
52	3.04795118	192.168.245.133	192.168.245.129	TLSv1.2	87	Encrypted Alert
53	3.04796266	192.168.245.129	192.168.245.133	DNS	76	Standard query 0x6f6f A temisleys.com
54	3.04803369	192.168.245.129	192.168.245.133	TCP	62	57657 -- 443 [ACK] Seq=1031 Ack=2023 Win=261632 Len=0
55	3.04806917	192.168.245.133	192.168.245.129	TCP	56	443 -- 57657 [FIN, ACK] Seq=30928 Ack=1031 Win=64128 Len=0
56	3.04813159	192.168.245.129	192.168.245.133	TCP	62	57657 -- 443 [ACK] Seq=1031 Ack=2024 Win=261632 Len=0
57	3.04817076	192.168.245.133	192.168.245.129	DNS	92	Standard query response 0x6f6f A temisleys.com A 192.168.245.133
58	3.04818707	192.168.245.129	192.168.245.133	TCP	60	57658 -- 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
59	3.04820813	192.168.245.133	192.168.245.129	TCP	60	443 -- 57660 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM=1 WS=128
60	3.04832253	192.168.245.129	192.168.245.133	TCP	62	57660 -- 443 [ACK] Seq=1 Ack=1 Win=262144 Len=0
61	3.04868848	192.168.245.129	192.168.245.133	TLSv1.2	237	Client Hello
62	3.04867577	192.168.245.133	192.168.245.129	TCP	56	443 -- 57660 [ACK] Seq=1 Ack=182 Win=64128 Len=0
63	3.05717411	192.168.245.133	192.168.245.129	TLSv1.2	1354	Server Hello, Certificate, Server Key Exchange, Server Hello Done
64	3.05739786	192.168.245.129	192.168.245.133	TCP	62	57660 -- 80 [ACK] Seq=182 Ack=1299 Win=268608 Len=0
65	3.06712064	192.168.245.129	192.168.245.133	TCP	60	57661 -- 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
66	3.06718278	192.168.245.133	192.168.245.129	TCP	60	80 -- 57661 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM=1 WS=128
67	3.06731551	192.168.245.129	192.168.245.133	TCP	62	57661 -- 80 [ACK] Seq=1 Ack=1 Win=262056 Len=0
68	3.06733094	192.168.245.133	192.168.245.129	HTTP	252	GET /msdownload/update/v3/static/trusted/en/authroots1.cab?78ce091359f3c0af HTTP/1.1
69	3.06739517	192.168.245.129	192.168.245.133	TCP	56	80 -- 57661 [ACK] Seq=1 Ack=197 Win=64128 Len=0
70	3.06842008	192.168.245.133	192.168.245.129	TCP	266	80 -- 57661 [PSH, ACK] Seq=1 Ack=197 Win=64128 Len=150 [TCP segment of a reassembled PDU]
71	3.06719783	192.168.245.133	192.168.245.129	HTTP	314	HTTP/1.1 200 OK (text/html)
72	3.06830951	192.168.245.129	192.168.245.133	TCP	62	57662 -- 80 [ACK] Seq=182 Ack=1299 Win=268608 Len=0

We receive notifications that volume shadow copy services were deleted upon running the sample.

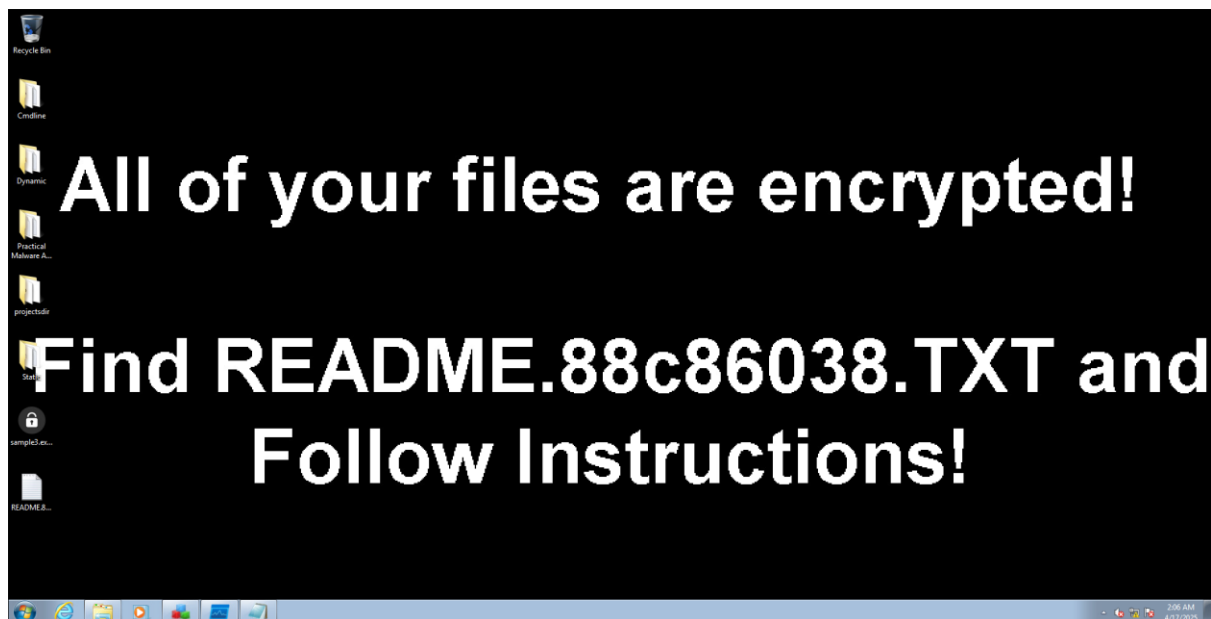


This makes sure that one is not able to restore their system after getting infected by the malware.

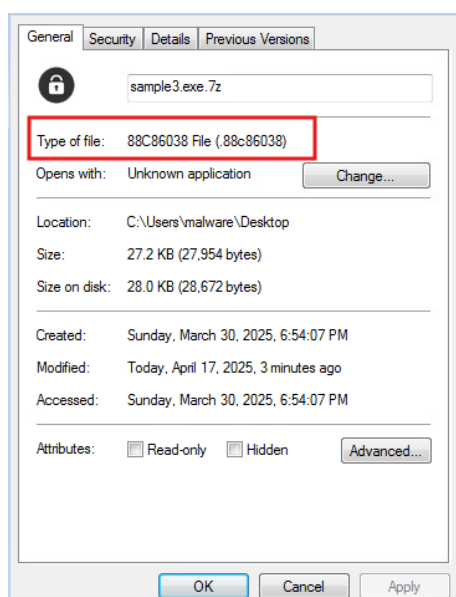
The desktop changes to the following when run on windows 10:



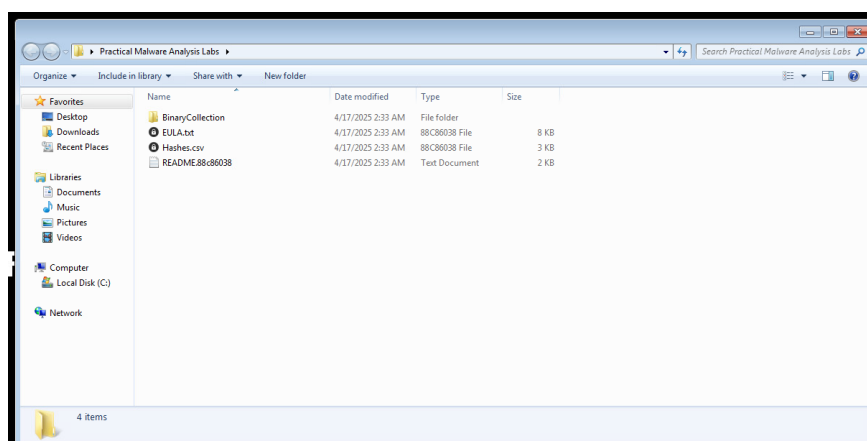
And on windows 7, it changes to this:



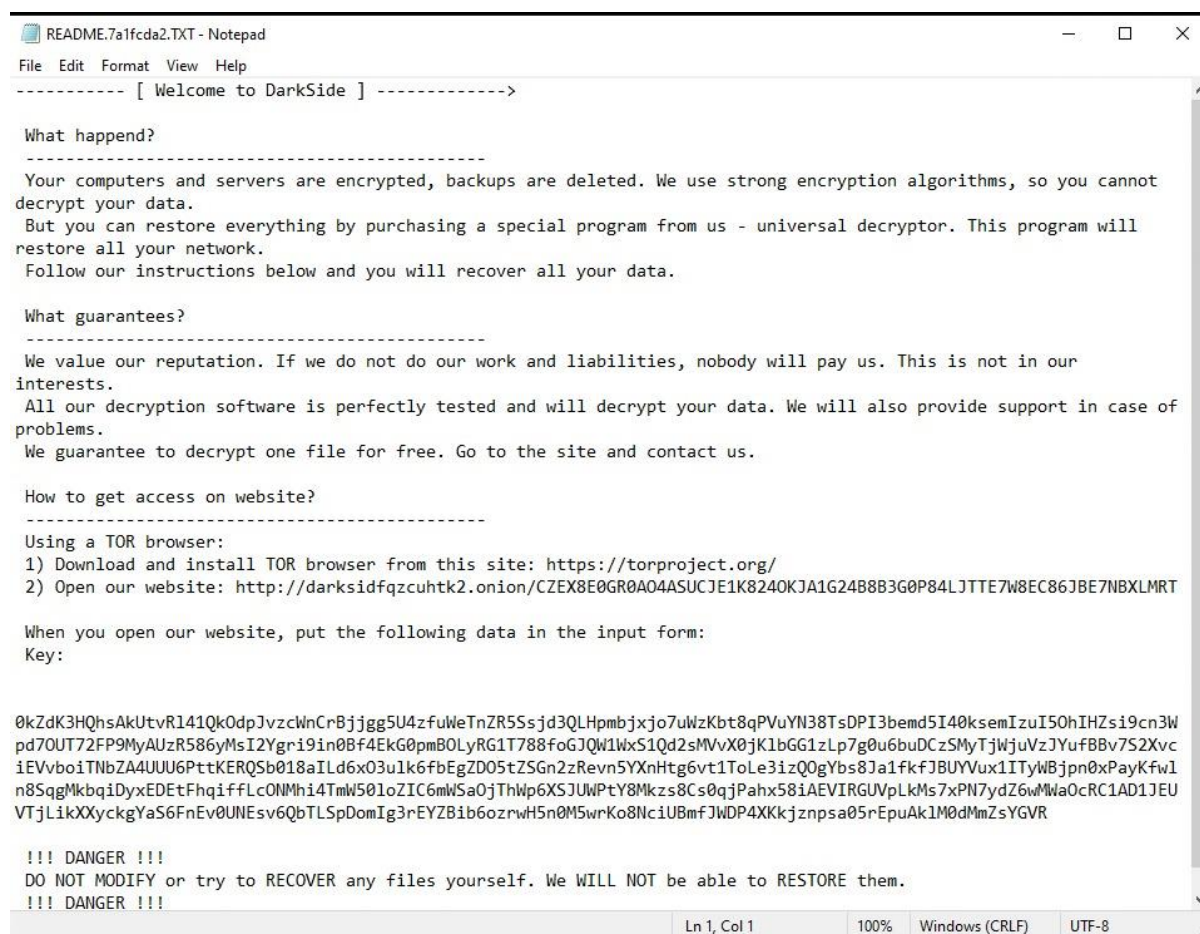
None of the files can then be used and have an extension 88c86038 (On win 7, and 7a1fcd2 on win 10).



All folders have a copy of the readme, and the files within them are similarly encrypted.



This is the readme file on windows 10:



```
README.7a1fcd2.TXT - Notepad
File Edit Format View Help
----- [ Welcome to DarkSide ] ----->

What happend?
-----
Your computers and servers are encrypted, backups are deleted. We use strong encryption algorithms, so you cannot
decrypt your data.
But you can restore everything by purchasing a special program from us - universal decryptor. This program will
restore all your network.
Follow our instructions below and you will recover all your data.

What guarantees?
-----
We value our reputation. If we do not do our work and liabilities, nobody will pay us. This is not in our
interests.
All our decryption software is perfectly tested and will decrypt your data. We will also provide support in case of
problems.
We guarantee to decrypt one file for free. Go to the site and contact us.

How to get access on website?
-----
Using a TOR browser:
1) Download and install TOR browser from this site: https://torproject.org/
2) Open our website: http://darksidfqzcuhtk2.onion/CZEX8E0GR0A04ASUCJE1K8240KJA1G24B8B3G0P84LJTTE7W8EC86JBE7NBXLMRT

When you open our website, put the following data in the input form:
Key:

0kZdK3HQhsAkUtvR141Qk0dpJvzcWnCrBjjgg5U4zfuWeTnZR5Ssjd3QLHpmbjxjo7uWzKbt8qPVuYN38TsDPI3bemd5I40ksemIzuI50hIHZsi9cn3W
pd70UT72FP9MyAUzR586yMsI2YgrI9in0Bf4EkG0pmB0LyRG1T788foGJQW1WxS1Qd2sMVvX0jK1bGG1zLp7g0u6buDCz5MyTjWjuVzJYufBBv7S2Xvc
iEVvboiTnbZA4UUU6PttKERQ5b018aILd6x03u1k6fbEgZD05tZSGn2zRevN5YXnHtg6vt1ToLe3izQ0gYbs8Ja1fkf7BUyVux1ITyWBjpn0xPayKfw1
n8SqmKkbqiDyxEDetFhqiffLcONMhi4TmW501oZIC6mWsa0jThWp6XSJUWpY8Mkzs8Cs0qjPahx58iAEVIRGUvPLkMs7xPN7ydZ6wMMwaOcRC1AD1JEU
VTjLkXxyckgYaS6FnEv0UNESv6QbTLSpDomIg3rEYZBib6ozrwH5n0M5wrKo8NciUBmfJWDP4XKkjznpsa05rEpuAk1M0dMmZsYGVr

!!! DANGER !!!
DO NOT MODIFY or try to RECOVER any files yourself. We WILL NOT be able to RESTORE them.
!!! DANGER !!!

Ln 1, Col 1    100%    Windows (CRLF)    UTF-8
```

The adversaries who go by the name “DarkSide” ask the victim to download tor to access the dark web from <https://torproject.org/> and then install the decrypting tool ‘universal decryptor’ from their tor website <http://darksidfqzcuhtk2.onion/>, and provide them the key written in the readme. They also apparently allow one free file decryption.

Indicators of compromise

1. The background change tells us that all our files are encrypted.
2. Unusable files.
3. Presence of a readme in all folders.
4. Inability to restore system (Disabling VSS).
5. DNS requests made to securebestapp20.com and temisleyes.com as seen in Wireshark.
6. Yara rule (also attached with submission):

```
remnux@remnux:~/Desktop$ cat detect_Darkside.yara
rule detect_Darkside
{
    meta:
        description = "Detects ransomware created by DarkSide"
        author = "Vandan Agrawal"
        date = "2025-04-16"

    strings:
        $vmpb = "VMProtect begin"
        $vmpe = "VMProtect end"
        $base = "ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz0123456789+/"
        $mz = { 4D 5A }

    condition:
        $mz at 0 and $vmpb and $vmpe and $base
}
remnux@remnux:~/Desktop$ yara -r detect_Darkside.yara .
detect Darkside ./sample3.exe
```